Efficacy And Safety Of Fixed Dose Combination Of Sitagliptin, Metformin And Glimepiride Tablets (Gemer-Sita®/Istamet-G®) In Indian Patients With Type 2 Diabetes Mellitus In Comparison With Co-administration Of Metformin And Glimepiride: Interim Analysis Of Phase 3, Randomized, Double-blind Study

Hemant Gupta[1], Rakesh Sahay[2], Vikas Agarwal[3], Sandesh Patil[4], A. Gopal Rao[5], Sandip Gofne[6], Ambana Gowda[7], Abhay Sahoo[8], Ramesh Duraisamy[9], Pradeep Rai[10], Raisa Shaikh[11], **Mandodari N Rajurkar[11*]**, Piyush M Patel[11#], Lalit K Lakhwani[11#], Suyog C Mehta[11], Sadhna J Joglekar[11#]

1 Grant Government Medical college and Sir JJ Group of Hospitals, Mumbai, India 2. Osmania General Hospital, Hyderabad 3. Surya Super speciality Hospital, Varanasi 4. Shree Siddhivinayak Maternity & Nursing Home, Nashik 5. Government Medical College & Government General Hospital, Srikakulam 6. District Civil Hospital, Aurangabad 7. Citizen hospital, Bangalore 8. IMS and SUM, Bhubaneswar 9 Kovai diabetes speciality hospital, Coimbatore 10. Opal Hospital Private Limited, Varanasi, 11. Sun Pharma Laboratories Limited, Mumbai, India
[#]Dr. Piyush Patel, Dr. Lalit K Lakhwani, and Dr. Sadhna Joglekar were full-time employees of Sun Pharma Laboratories Limited during conduct of the study.

*Presenting Author

Paper presented at the 7th International Diabetes Summit, 10th March 2023, Pune, MS, India

Sitagliptin Plus Metformin Plus Glimepiride – Unmet Need & Rationale

- Among **Indian Patients with diabetes**, mean HbA1c is 8.9% (Diabcare India Study – 2011) & 2/3rd are not at target HbA1c (ICMR-INDIAB Study)
- With over 30% of patients taking 3-4 tablets/day, pill burden results in poor treatment adherence, which in turn leads to inadequate glycemic control
- In a **meta-analysis involving data from 70,573 patients**, use of FDCs with oral anti diabetic agents was associated with lower HbA1c and higher medication possession ratio compared to co-administered dual therapy use in type II DM
- **Studies conducted with Sitagliptin & Metformin** (100mg/1000mg) and up-titrated to 100/2000 dose – in case of patients with baseline HbA1c in range of 7.5% – 12% have demonstrated 40% – 50% of patients are not reaching targeted HbA1c
- **Triple drug therapy** with Sitagliptin + Metformin + Sulfonyl Urea has shown to further reduce HbA1c by 0.6 to 0.8%

1. Unnikrishnan R, et al. Nat Rev Endocrinol 2016; 12: 357 – 370; 4. 2. V. Mohan, et al. Diabet Indian J Med Res 125, March 2007, pp 217-230; 3. Hans S, et al Curr Med Res Opin 2012 Jun; 28(6): 967-77; 4. Reasner C, et al. Diabetes, Obesity and Metabolism 2011; 13: 644 – 52; 5. Perez-Monteverde A, et al. Int J Clin Pract Sept 2011; 65(9): 930 – 38; 6. Wainstein J, et al. Diabetes, Obesity and Metabolism 2012; 14(5):409-18; 7. Williams-Herman D, et al. Diabetes, Obesity and Metabolism 2010; 12: 442-51

Sitagliptin Plus Metformin Plus Glimepiride – Therapeutic Justification

- **UK & EMEA prescribing information of Sitagliptin plus metformin** recommends it in combination sulphonylurea (i.e., triple combination therapy) as an adjunct to diet and exercise in patients inadequately controlled on their maximal tolerated dose of metformin and a sulphonylurea
- **ADA Guidelines** recommend addition of DPP4 to SU therapy wherein cost is a concern and HbA1c is not on target
- **NICE guidelines 2017 recommends** second intensification with triple therapy of DPP4+Metformin+SU if patients with combination therapy of DPP4 & Metformin have an HbA1c >7.5%
- In Indian context – near 40-50% of patients receiving Sitagliptin also receive Glimepiride and Metformin*
- **Real world evidence studies demonstrate the Hazard ratio for hypoglycemia to be lower** with a triple drug therapy of **SU+DPP4+Metformin** compared to triple drug therapy with **SU + Glitazones + Metformin – 5.07 v/s 6.32**
- **Addition of Sitagliptin to ongoing glimepiride plus metformin therapy has shown to have reduced incidence of hypoglycemia** as compared to vice versa
- **Addition of Sitagliptin helps to reduce dose of glimepiride and is shown to preserve the β-cell function** – seen through studies evaluating PI/I ratio (lower PI/I ratio)

8. Janumet – UK Prescribing information accessed on 31st July 2020; 9. ADA Guidelines 2020;10. NICE Guidelines for Diabetes Mellitus Version 2017 accessed on 31st July 2020; 11. Hippisley-Cox J et al. BMJ 2016;352:i450

*SMSRC data Nov-Feb 2020

Rationale of lower dose of Glimepiride in combination with Sitagliptin and Metformin

Glimepiride high dose →

- Progressive β cell dysfunction
- Higher risk of Hypoglycemia events and weight gain
- Efficacy plateau in uncontrolled patients

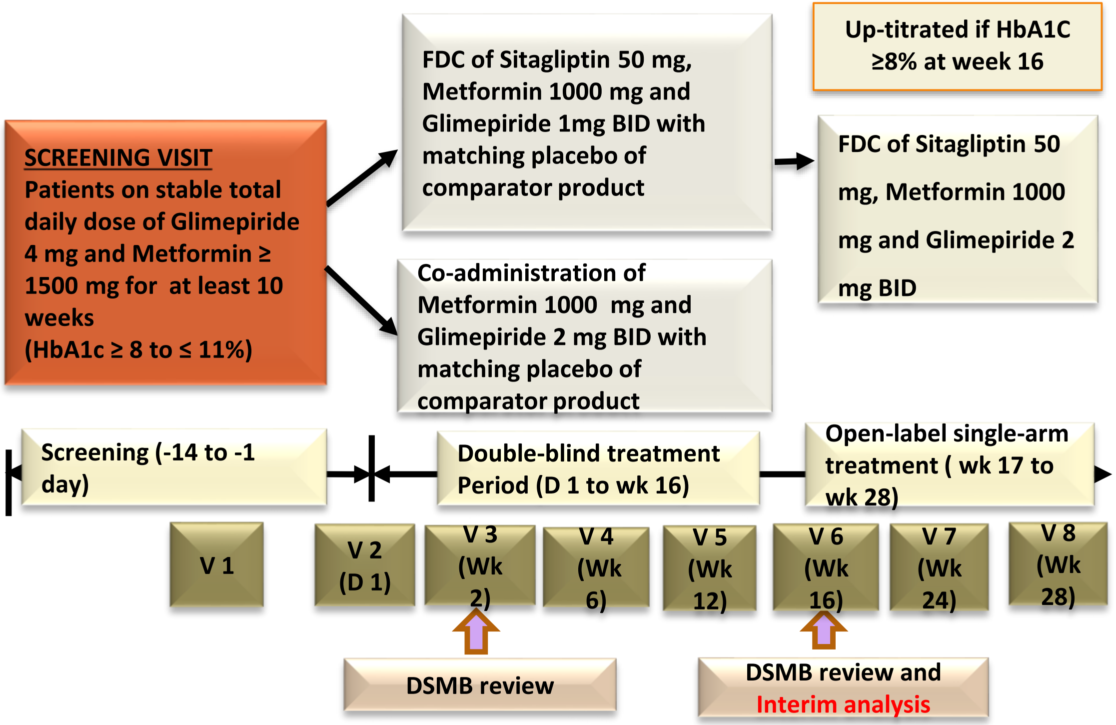
Sitagliptin (DPP-4) addition to Lower dose Glimepiride and Metformin:

- The prolonged activity of intact GLP-1 by DPP-4 inhibition is likely to both improve glucose-induced insulin secretion and to decrease glucagon levels, resulting in the maintenance of β-cell function
- Reduced Hypoglycemia events and is weight neutral
- Provides a complementary mechanism for glucose induced insulin secretion leading to improved diabetes control
- Sitagliptin exhibits known synergism with Metformin through several mechanisms

β cell = beta cell; DPP-4 = dipeptidyl peptidase-4; GLP = Glucagon-like peptide 1.

Sato A et al. J Clin Med Res 2019; 11(1): 15 – 20

Objective and Study design



Objective (Interim Analysis)

Efficacy and safety of FDC of Sitagliptin, Metformin and Glimepiride Tablets (50 mg/1000 mg/1 mg) and (50 mg/1000 mg/2 mg) given BID in comparison to Co-administration of Metformin Hydrochloride 1000 mg and Glimepiride 2 mg BID till Week 16 in 127 patients.

-

BID = twice daily; D=day; DSMB = Data Safety Monitoring Board; FDC = fixed dose combination; HbA1c = Glycosylated hemoglobin, Wk=week.

Study Endpoints

Primary Endpoint:

Efficacy:

Mean Change in HbA1c from baseline at the end of Week 16

Secondary Endpoint:

Efficacy:

1. Mean Change in HbA1c from baseline at the end of Week 28
2. Mean PPBG from baseline at the end of Weeks 12, 16, 24 and 28
3. Mean Change in FBG from baseline at the end of Weeks 12, 16, 24 and 28
4. Proportion of participants achieving HbA1c < 7.0% at Weeks 12, 16 and 28
5. Number of patients requiring hypoglycemia management
6. Number of patients requiring rescue medications

Safety

1. Safety assessment includes TEAEs assessment during the study

Study Hypothesis and Sample Size Estimation

Study Hypothesis:

Superiority in efficacy and comparable safety of Triple FDC over dual combinations.

Sample Size Estimation

- Assuming 85% power and 5% alpha error with a standard deviation of 1, sample size was calculated to detect a treatment difference of 0.4%
- Assuming 15% attrition, 134 patients in each arm were needed to be enrolled
- Total sample of 402 patients was found to be satisfactory to meet the study objectives

Statistical analysis

- Of 392 planned patients, 127 patients were planned to be randomized initially for interim analysis.
- Interim analysis was performed at Week 16
- All statistical analysis was done using SAS version 9.4
- Continuous variables were presented using mean, standard deviation (SD), minimum and maximum while categorical variables were presented using counts and percentages
- Baseline Characteristics were compared using two sample t-test
- *Efficacy endpoints:*
  - Mean change in HbA1c/ FBG/ PPBG from baseline to Week 16 were evaluated using ANCOVA model with treatment group variables as fixed factors and baseline HbA1c value as covariate
  - Proportion of patients achieving HbA1c < 7.0% at Week 12 and Week 16 were evaluated using Chi-square test
- *Safety endpoints:* safety parameters were evaluated using paired t-test for within group comparison and two sample t-test for between group comparison
- All statistical tests were conducted at 5% significance and 95% confidence interval

Disposition of Patients

Total 161 patients were screened; out of which, 127 patients were randomized.

	FDC of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets (50 mg/1000 mg/1 mg) BID (Test) (N = 60)	Co-administration of Metformin Hydrochloride 500 mg (2 tablets BID) and Glimepiride tablets 2 mg (BID) (Comparator) (N = 67)	Overall (N = 127)
ITT Population	60 (100%)	67 (100%)	127 (100%)
PP Population	55 (91.7%)	63 (94.0%)	118 (92.9%)
Safety Population	60 (100%)	67 (100%)	127 (100%)
mITT Population	60 (100%)	67 (100%)	127 (100%)

BID: Twice Daily; **FDC**: Fixed Dose Combination; **ITT**: Intention-To-Treat; **mITT**: modified Intention-To-Treat; **N**: Number of patients; **PP**: Per Protocol

Demographic and Baseline characteristics (Safety Population)

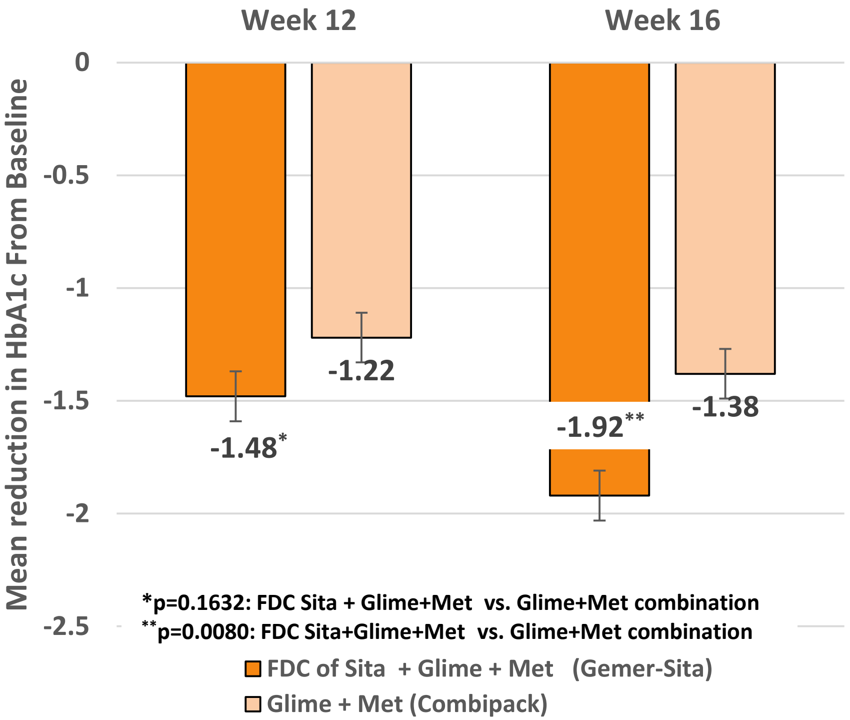
		FDC of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets (50 mg/1000 mg/1 mg) BID (Test) (N = 60)	Co-administration of Metformin Hydrochloride 500 mg (2 tablets BID) and Glimepiride tablets 2 mg (BID) (Comparator) (N = 67)	Test Vs Comparator
Visits	**Statistic Summary**	**Actual**	**Actual**	***p value***
Gender n(%)	Male	31 (51.7%)	37 (55.2%)	0.6882
	Female	29 (48.3%)	30 (44.8%)	
Age (years)	Mean ± SD	51.70± 8.44	49.63±10.21	0.2176
	Min, Max	24.00, 64.00	26.00, 65.00	
Height (cms)	Mean ± SD	161.4±6.64	162.4±6.83	0.3939
	Min, Max	148.00, 174.00	148.00, 175.00	
Weight (kg)	Mean ± SD	66.35±10.08	69.95±9.88	0.0440
	Min, Max	47.70, 88.00	49.30, 101.80	
BMI (Kg/m²)	Mean ± SD	25.41±3.16	26.56±3.80	0.0687
	Min, Max	17.10, 32.00	18.90, 40.70	

No significant difference in baseline characteristics between both arms except for body weight

Interim Efficacy Result (Primary Endpoint): Reduction in HbA1c from Baseline to Week 12 and Week 16

		FDC of Sita, Met and Glime tablets (50 mg/1000 mg/1 mg) BID (Test) (N = 60)		Co-administration of Met 500 mg (2 tablets BID) and Glime tablets 2 mg (BID) (Comparator) (N=67)	
Visits	Statistic Summary	Actual	Change from Baseline	Actual	Change from Baseline
Baseline (Visit 2)	n	60		67	
	Mean ± SD	9.18±0.86		9.13±0.90	
	p-value				
Week 12 (Visit 5)	n	59	59	66	66
	Mean ± SD	7.72± 1.11	-1.48±1.06	7.92±1.10	-1.22±0.98
	p-value		<.0001		<.0001
Week 16 (Visit 6)	n	59	59	66	66
	Mean ± SD	7.28± 1.07	-1.92± 1.03	7.76± 1.24	-1.38± 1.20
	p-value		<.0001		<.0001

Reduction in HbA1c from Baseline to Week 12 and Week 16

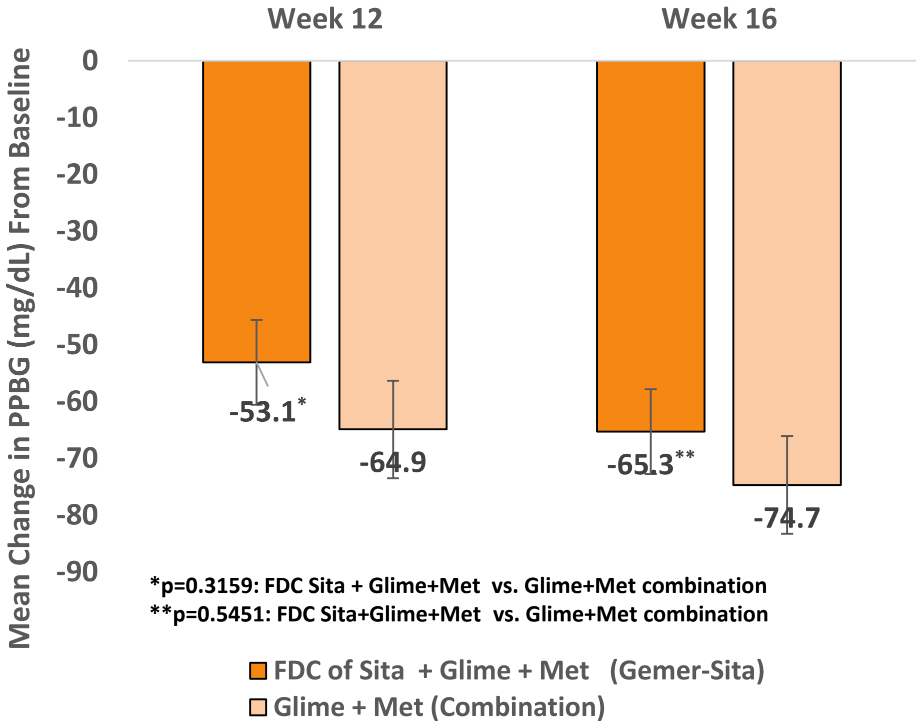


HbA1c=glycated hemoglobin.

Interim Efficacy Result (Secondary Endpoint): Reduction PPBG from Baseline to Week 12 and Week 16

		FDC of Sita, Met and Glime tablets (50 mg/1000 mg/1 mg) BID (Test) (N = 60)		Co-administration of Met 500 mg (2 tablets BID) and Glime tablets 2 mg (BID) (Comparator) (N=67)	
Visits	Statistic Summary	Actual	Change from Baseline	Actual	Change from Baseline
Baseline (Visit 2)	n	60		67	
	Mean ± SD	265.6 ± 57.61		280.2 ± 70.44	
Week 12 (Visit 5)	n	59	59	66	66
	Mean ± SD	210.2± 51.11	-53.1±60.17	216.0±51.62	-64.9± 69.06
	p-value		<.0001		<.0001
Week 16 (Visit 6)	n	59	59	66	66
	Mean ± SD	198.0± 42.89	-65.3±56.33	206.2±43.01	-74.7± 71.55
	p-value		<.0001		<.0001

Mean Reduction in PPBG (mg/dL) from Baseline at the End of Week 12 & Week 16



*p=0.3159: FDC Sita + Glime+Met vs. Glime+Met combination
**p=0.5451: FDC Sita+Glime+Met vs. Glime+Met combination

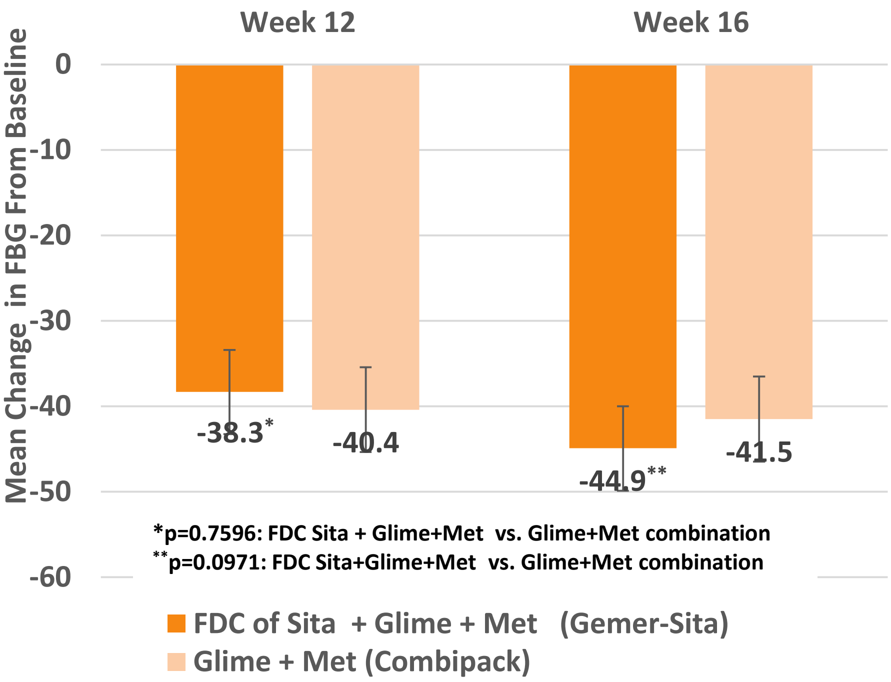
Paper presented at the 7th International Diabetes Summit, 10th March 2023, Pune, MS, India

Interim Efficacy Results (Secondary Endpoint): Reduction in FBG from Baseline to Week 12 and Week 16

		FDC of Sita, Met and Glime tablets (50 mg/1000 mg/1 mg) BID (Test) (N = 60)		Co-administration of Met 500 mg (2 tablets BID) and Glime tablets 2 mg (BID) (Comparator) (N=67)	
Visits	Statistic Summary	Actual	Change from Baseline	Actual	Change from Baseline
Baseline (Visit 2)	n	60		67	
	Mean ± SD	183.0±38.48		192.5±40.16	
Week 12 (Visit 5)	n	59	59	66	66
	Mean ± SD	143.8± 42.03	-38.3± 38.78	151.9± 37.95	-40.4± 37.83
	p-value		<.0001		<.0001
Week 16 (Visit 6)	n	59	59	66	66
	Mean ± SD	137.2± 33.51	-44.9± 35.51	150.8± 34.98	-41.5±35.14
	p-value		<.0001		<.0001

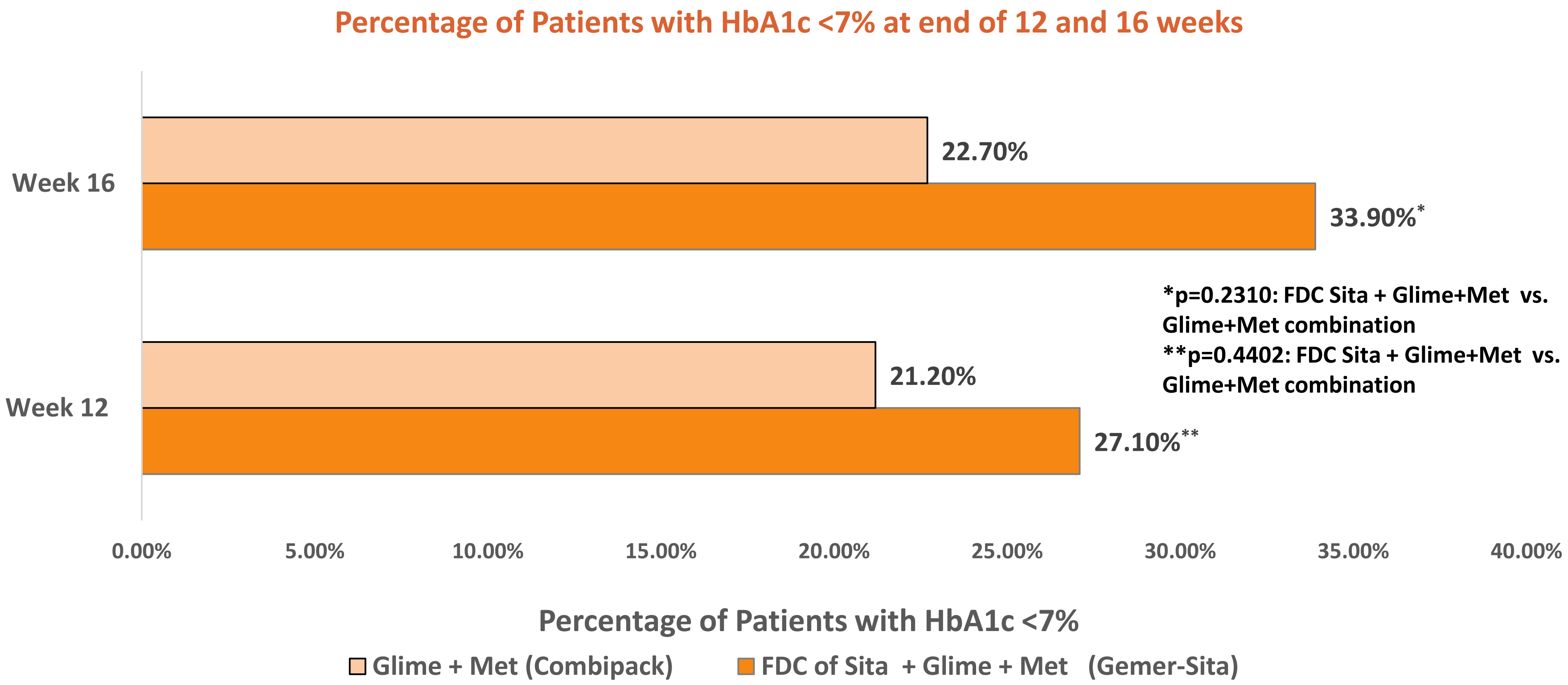
FBG=Fasting blood glucose;

Mean Reduction in FBG (mg/dL) from Baseline at the End of Week 12 & Week 16



*p=0.7596: FDC Sita + Glime+Met vs. Glime+Met combination
**p=0.0971: FDC Sita+Glime+Met vs. Glime+Met combination

Interim Efficacy Results (Secondary Endpoint): Proportion of Patients Achieving HbA1c <7.0% at Week 12 and Week 16



Interim Safety Results (Till Week 16)

	Test: FDC of sitagliptin Phosphate, metformin Hydrochloride and glimepiride tablets (50 mg/1000 mg/1 mg) BID (N=60)		Comparator 1: Co-administration of metformin hydrochloride 500 mg (2 tablets BID) and glimepiride tablets 2 mg (BID) (Comparator) (N=67)		Overall (N=127)	
	Patients n (%)	Events E	Patients n (%) [1]	Events E	Patients n (%) [1]	Events E
Any AEs	12 (20.0%)	15	11 (16.4%)	14	23 (18.1%)	29
Any TEAEs	12 (20.0%)	15	11 (16.4%)	14	23 (18.1%)	29
Drug-related TEAEs	5 (8.3%)	5	3 (4.5%)	3	8 (6.3%)	8

- **All the AEs were mild in nature**
- **No SAEs, deaths, severe or life-threatening TEAEs reported during the study**
- **The action taken for AEs was 'Dose not Changed' for 14 events, and 'Not applicable' for 15 events**
- **Out of 29 adverse events, seven events were reported 'Probable/likely', one event as 'Possible' and 21 as 'Unlikely'**
- **The outcome status for all the AEs were recovered.**
- **No TEAEs required permanent treatment discontinuation.**
- **Total 7 hypoglycemia events were observed during the study: 3 patients (5 events) from Test arm and 1 patient (2 events) from Comparator arm**
- **All the hypoglycemia events were level 1 and asymptomatic**
- **None of them required management**
- **None of the patient required rescue medications during the study**
- AE=adverse events; BID=twice daily; FDC=fixed dose combination; SAEs=serious adverse events; TEAEs=treatment-emergent adverse events.

Safety Results till Week 16– AEs by SOC and PT

SUN PHARMA

Treatment Emergent Adverse Events by System Organ Class (SOC) and Preferred Term (PT) - Safety Population

System Organ Class Preferred Term	FDC of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets (50 mg/1000 mg/1 mg) BID (Test) (N=60)		Co-administration of Metformin Hydrochloride 500 mg (2 tablets BID) and Glimepiride tablets 2 mg (BID) (Comparator) (N=67)		Overall (N=127)	
	n(%)	E	n(%)	E	n(%)	E
All TEAEs	12 (20.0%)	15	11 (16.4%)	14	23 (18.1%)	29
Gastrointestinal disorders	**4 (6.7%)**	**5**	**8 (11.9%)**	**10**	**12 (9.4%)**	**15**
Abdominal pain upper	1 (1.7%)	1	1 (1.5%)	1	2 (1.6%)	2
Diarrhoea	1 (1.7%)	1	2 (3.0%)	3	3 (2.4%)	4
Hyperchlorhydria	0	0	1 (1.5%)	1	1 (0.8%)	1
Nausea	0	0	3 (4.5%)	3	3 (2.4%)	3
Vomiting	2 (3.3%)	3	2 (3.0%)	2	4 (3.1%)	5
Metabolism and nutrition disorders	**3 (5.0%)**	**5**	**1 (1.5%)**	**2**	**4 (3.1%)**	**7**
Hypoglycaemia	3 (5.0%)	5	1 (1.5%)	2	4 (3.1%)	7
Musculoskeletal and connective tissue disorders	**1 (1.7%)**	**1**	**0**	**0**	**1 (0.8%)**	**1**
Arthralgia	1 (1.7%)	1	0	0	1 (0.8%)	1
Nervous system disorders	**3 (5.0%)**	**3**	**2 (3.0%)**	**2**	**5 (3.9%)**	**5**
Headache	3 (5.0%)	3	2 (3.0%)	2	5 (3.9%)	5
Respiratory, thoracic and mediastinal disorders	**1 (1.7%)**	**1**	**0**	**0**	**1 (0.8%)**	**1**
Rhinorrhoea	1 (1.7%)	1	0	0	1 (0.8%)	1

BID: Twice daily; E: Number of Events; FDC: Fixed dose combination; N: Number of subjects

[1] Body System totals are not necessarily the sum of the individual adverse events since a patient may have reported more than one adverse events in the same body system; [2] Adverse events are coded into system organ class and preferred term using MedDRA version 24.0; [3] Percentages are computed using N provided in the Column header; [4] If AE is with same Start date but different severity then worst case or latest (whichever applicable) will be consider.

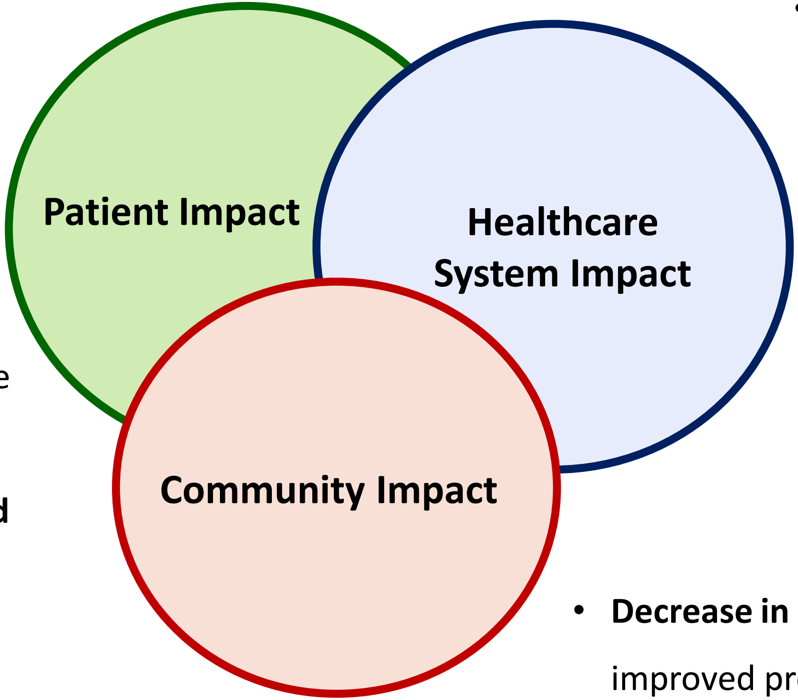
Study Conclusion

Triple Fixed dose combination of sitagliptin, metformin, glimepiride was significantly superior to co administered metformin and glimepiride in reducing HbA1c at Week-16. Study medications were safe and well tolerated.

Fixed dose combination will provide a better therapeutic alternative by limiting the rate of beta-cell exhaustion for patients on high dose sulfonylurea.

Sitagliptin, Metformin and Glimepiride – Impact across Value Chain of T2DM Management

- **Better HbA1c control** at lower dose of Glimepiride
- **Less deterioration of pancreatic β cell function** – through β cell preservation achieved due to lower glimepiride dose
- **Benefit without increased risk** of Hypoglycaemia
- Improved compliance



- **Decrease in healthcare resource utilization and strain on healthcare** system by slowing disease progression (by β cell exhaustion) , more patients achieving target HbA1c and thereby reducing/delaying long-term complications

- **Decrease in economic impact**, improved productivity of society and better quality of life

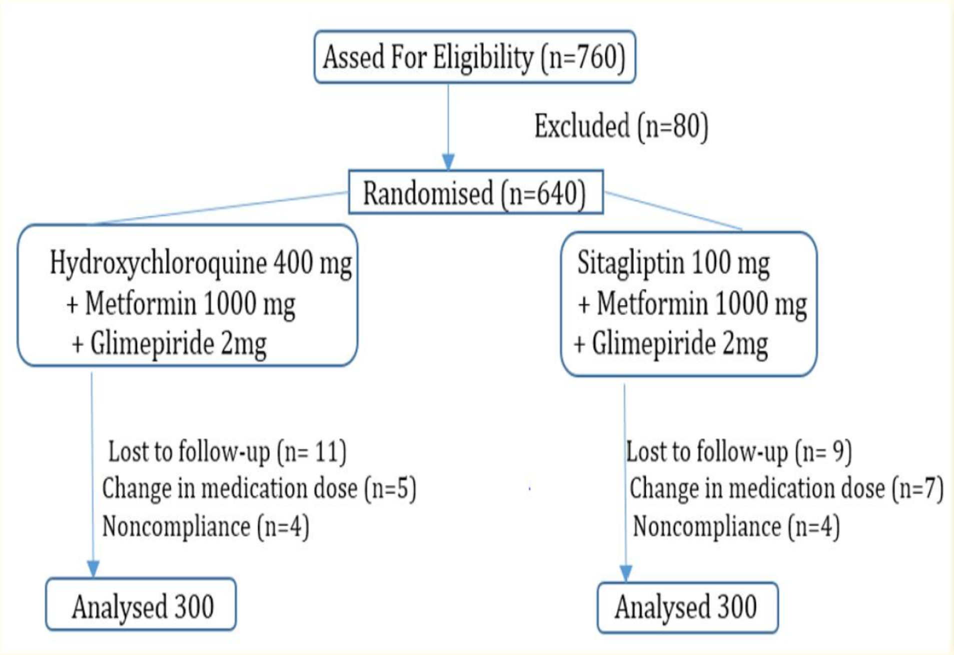
β cell = beta cell; HbA1c = Glycosylated hemoglobin, T2DM = Type2 diabetes mellitus.

Thank You

Back up slides

Triple drug therapy – Sitagliptin vs Hydroxychloroquine + Glimepiride & Metformin in T2DM

- **Comparative Observational Study conducted across 7 centers in India involving 600 patients with T2DM**
- **T2DM Patients of either sex, aged 38 – 65 years, average BMI 29.5 kg/m2 with h/o disease from 2 years**



- Reduction in HbA1c – 1.2 in Sita arm vs 1.4 in HCQ arm
- **Incidence of hypoglycemia was similar among groups** (Incidence of hypoglycemia reported with HCQ 400mg + Glimepiride 4mg + Metformin 500mg is mild and 2.1% in published studies[13])
- **No episode of hypoglycemia exhibited marked severity** (i.e., loss of consciousness or requirement for medical assistance)

Singh UP et al. EC Endocrinology and Metabolic Research 3.4 (2018): 147-155; 13. Charavarthy HN et al. J of Endocrinol Invest 2021; 44:481-92

Summary

- With assumed mean difference of 0.3% HbA1c with FDC of Sita+Met+Glim (50 mg/1000 mg/1 mg) BID (test) in comparison to Met+Glim (1000 mg/2 mg) BID (comparator), total 392 patients was targeted in the study.
- In comparison to our assumption of 0.3% HbA1c difference, we have **observed the difference of 0.54% HbA1c between two arms** (p = 0.008 between groups)
- The difference has reached statistically significant superiority of test against comparator from data of 127 patients
- These results are expected to have **significant implications in a real world setting** by allowing decreasing dose of glimepiride yet lead to a significantly more reduction in HbA1c and allowing more patients to not only reduce pill burden but also achieve target HbA1c

Summary

- Continuing the patients on high dose sulfonylurea in comparator arm does not seem righteous for remaining study patients
- **Considering the above rationale, we request you to kindly grant us approval to manufacture Fixed Dose Combination of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets**

Study Objectives

Primary Objective:

To assess the efficacy of FDC of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets (50 mg/1000 mg/1 mg) given twice daily (BID) in comparison to Co-administration of Metformin Hydrochloride 500 mg (2 tablets given BID) and Glimepiride 2 mg tablets given BID in patients treated for type 2 diabetes mellitus

Secondary Objective:

- To assess the safety of FDC of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets (50 mg/1000 mg/1 mg) given BID in comparison to Co-administration of Metformin Hydrochloride 500 mg (2 tablets given BID) and Glimepiride 2 mg tablets given BID in patients treated for type 2 diabetes mellitus
- To assess the efficacy and safety of FDC of Sitagliptin Phosphate, Metformin Hydrochloride and Glimepiride tablets (50 mg/1000 mg/2 mg) given BID in patients treated for type 2 diabetes mellitus

Study 1: Sitagliptin plus baseline low dose Glimepiride vs baseline only high dose Glimepiride

Open labelled comparative study to evaluate the effects of combination therapy of Sitagliptin 50mg/day + Glimepiride 1mg/day in patients receiving 2 – 6mg/day of Glimepiride (n = 26)

Key Criteria

- Patients with poorly controlled Type-2 DM
- Currently taking glimepiride (≥ 2mg/day)

Glimepiride 2-3 mg/day (low dose group) → **Sitagliptin 50mg/day + Glimepiride 1mg/day**

Glimepiride 4-6 mg/day (high dose group) → **Sitagliptin 50mg/day + Glimepiride 1mg/day**

Endpoints

- Change in HbA1c from baseline to 24 and 48 weeks
- Safety assessment with focus on hypoglycaemia over treatment period

Ishii H et al. J Clin Med Res 2014; 6(2): 127 – 132

Study 1: Sitagliptin plus low dose Glimepiride vs high dose Glimepiride

Efficacy

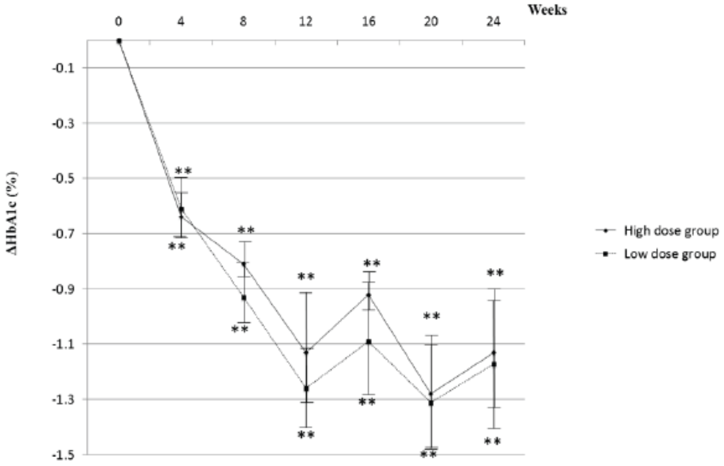


Figure 3. Degrees of HbA1c reduction are shown. A statistical analysis was performed by the Wilcoxson rank sum test, paired t-test.* *P < 0.01 vs. 0 week.

In patients with baseline glimepiride 2-3 mg/day or 4-6mg/day – Sitagliptin plus use of low dose glimepiride (dose reduction) results in significant and comparable reduction in HbA1c (-1.1 to 1.2%) in both treatment arms

Safety

- None of the patients showed even mild hypoglycemic symptoms, such as palpitation, sweating, or unusual feelings of hunger, at each visit
- Body weight remained unchanged in both groups during the study period

Ishii H et al. J Clin Med Res 2014; 6(2): 127 – 132

Study 2: Sitagliptin plus low dose Glimepiride vs high dose Glimepiride

SUN PHARMA

Prospective randomized, multicenter, open-label, comparative study was performed to analyze the effects of Sitagliptin on glycemic control and maintenance of beta-cell function in patients with poorly controlled type 2 diabetes treated with low-dose glimepiride (n = 41)

Key Criteria

- Patients with T2DM aged 20 – 80 years
- HbA1c – 7.4 to 9.4
- Treated with glimepiride ≤ 2mg/day

Randomized 1:1

Glimepiride 6 mg/day (Glimepiride group) [gradual increase] (n = 19)

Sitagliptin 50mg/day plus Glimepiride ≤2mg/day (Sitagliptin group) (n = 22)

Endpoints

Primary

- C-peptide immunoreactivity
- Proinsulin/Insulin (P/I) Ratio

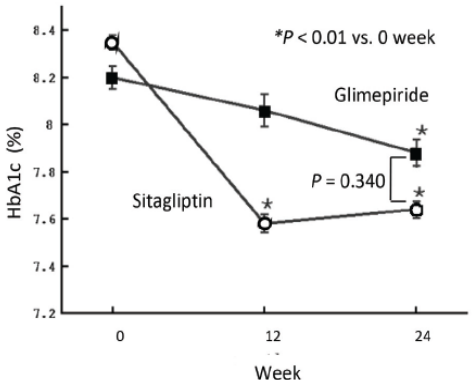
Key Secondary

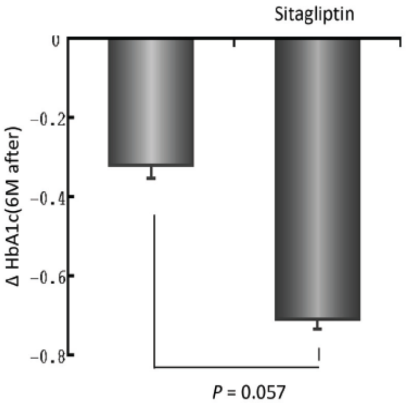
- Differences in HbA1c (target <6.9%),
- fasting plasma glucose
- Homeostatic model assessment of β-cell function (HOMA-β)

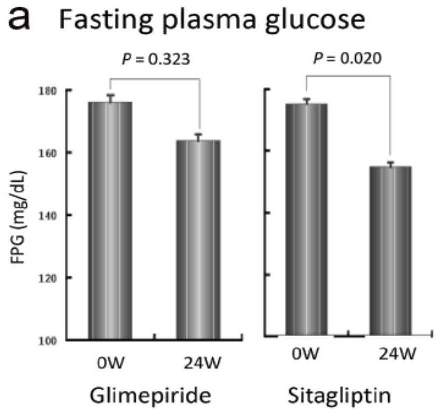
Sato A et al. J Clin Med Res 2019; 11(1): 15 – 20

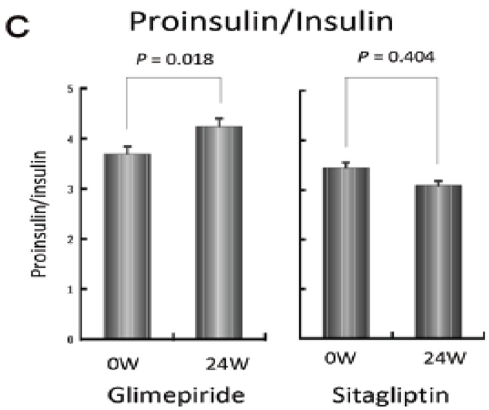
Protection of Pancreatic Beta cells Study: High dose Glimepiride vs 'Low dose Glimepiride + Sitagliptin'

Efficacy in Treatment Arms







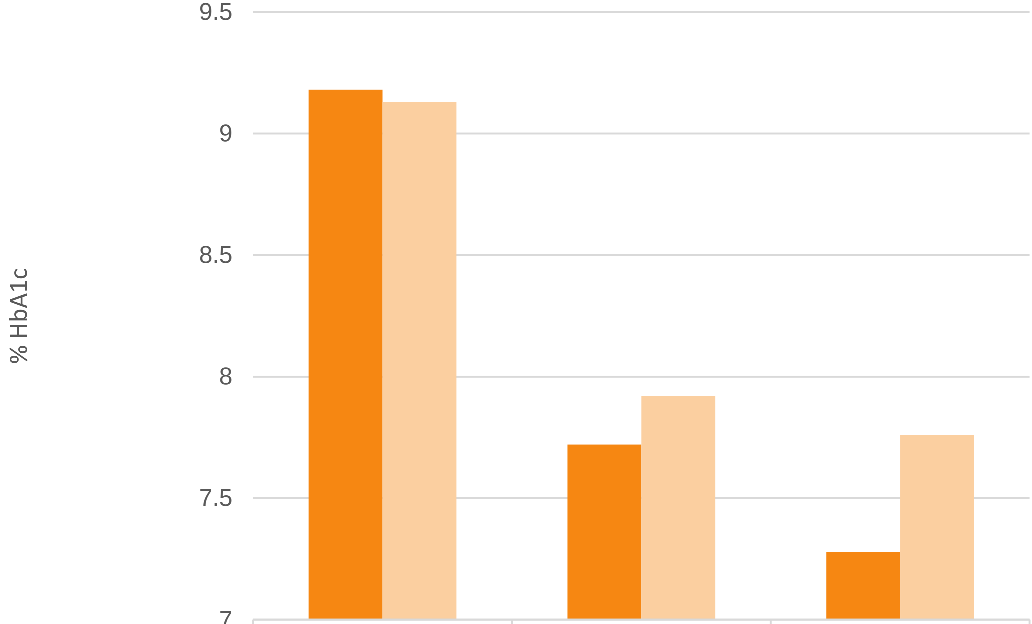


- **Significant higher reduction in HbA1c at 12 weeks** with Sita + Glime vs Glime high dose
- **Near significant reduction in 6 month HbA1c** with Sita plus Glime vs High dose Glime
- **Target fasting plasma glucose** – achieved in **36.7% of Sitagliptin group vs 16.7% in high dose glimepiride** group
- Glimepiride high dose group→ **Higher Pro-insulin/Insulin ratio** indicating towards **Beta cell dysfunction**
- Low dose Glimepiride plus Sitagliptin **preserves Beta cell function (lowered PI/I)**
- **Hypoglycaemia was mild and comparable in both arms (1 patient in glimepiride and 2 patients in Sita arm)**

Sato A et al. J Clin Med Res 2019; 11(1): 15 – 20

Interim Efficacy Result: Change in HbA1c from Baseline to Week 12 and Week 16

%HbA1c (Modified Intention To Treat Population)



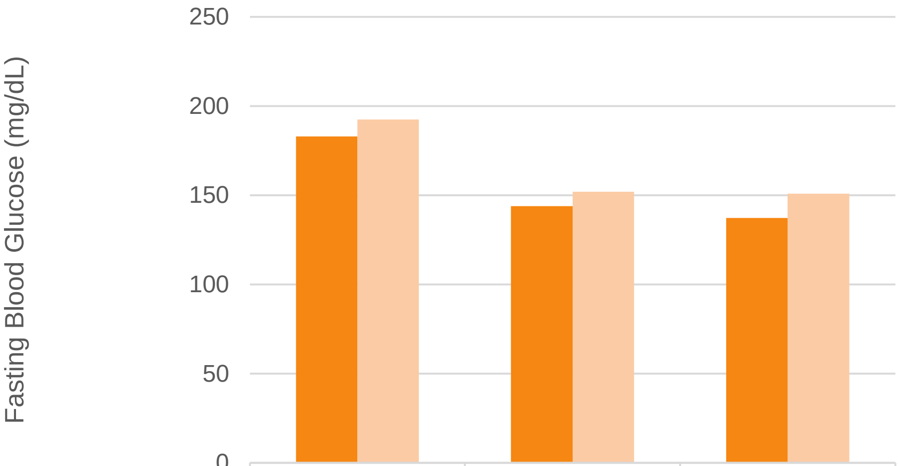
	Baseline	Week 12	Week 16
Test (N=60)	9.18	7.72	7.28
Comparator (N=67)	9.13	7.92	7.76

- **Statistically significant reduction in HbA1c in test arm compared to comparator arm was observed at Week 16 (p=0.008).**
- **Reduction in HbA1c at Week 12 was comparable between test and comparator arms (p=0.1632).**
- **Similar results were observed in intention to treat population and per protocol population (N= 55 in Test arm and N=63 in Comparator arm).**

HbA1c=glycated hemoglobin.

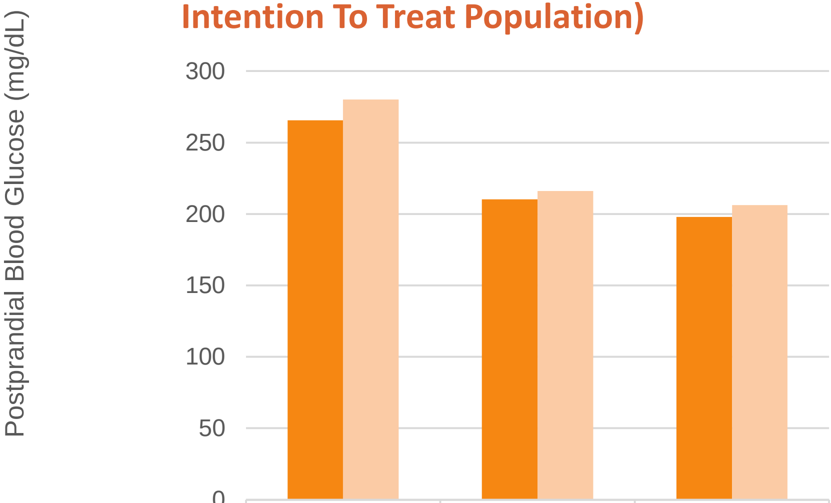
Interim Efficacy Results: Change in FBG and PPBG from Baseline to Week 12 and Week 16

Fasting Blood Glucose (Modified Intention To Treat Population)



	Baseline	Week 12	Week 16
Test (N=60)	183	143.8	137.2
Comparator (N=67)	192.5	151.9	150.8

Postprandial Blood Glucose (Modified Intention To Treat Population)



	Baseline	Week 12	Week 16
Test (N=60)	265.6	210.2	198
Comparator (N-67)	280.2	216	206.2

Reduction in FBG and PPBG from baseline to week 12 and week 16 was statistically significant within treatment arms ($p<0.001$, each). Reduction in FBG was comparable between the arms at Week 12 ($p=0.7596$) and Week 16 ($p=0.0971$). Likewise, Reduction in PPBG was comparable between the arms at Week 12 ($p=0.3159$) and Week 16 ($p=0.5451$). Similar results were observed in ITT population and PP population.

FBG=Fasting blood glucose; PPBG=Post prandial blood glucose.